Contact

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Skills

Generative AI: Prompt Engineering, Hugging Face, Transformers
Deployment, Containerisation
Docker, TailScale, SkyPilot, Flask Streamlit,
DBMS:
MySQL, PostgreSQL, MongoDB, MariaDB
Analytics & Visualisation:
Python, Power BI, MS-Excel
Data Science Libraries:
Pytorch, Keras, sklearn, pandas,
Programming: Data Versioning, HTML,CSS,

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A Case Study Approach ,
Crash Course on Python

Teaching

Teaching Assistant, NPTEL
(Jan 2024- April 2024)
noc24-cs47:

The Joy of Computing Using Python
Prepared weekly assignments & handled the forum for the course having around 60,000 registered learners.

Student Mentor, IITM BS Degree Programme
(Mar 2024- April 2024)
BSMA1004:

Statistics for Data Science 2.
Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of 25 students

Volunteering

Agentic AI Workshop Series by Prof Raghav Kulkarni, CMI |
August 2026 -Present

Conducting Sessions every Sunday from 4PM to 7 PM on Agentic AI

Maria Thurkadayil

DATA SCIENTIST

Summary

Master's Degree Student building skills in the field of Data Science, Looking for opportunities to integrate research with latest technologies

Experience

Research Intern , IIT Jammu | June 2026 - Present

Quality-Aware Natural Language Data Curation Pipeline (Ongoing)
- Worked on a high-dimensional NLP clustering infrastructure (sentence-transformers) for text style and quality assessment; extended the core framework by implementing a custom Iterative Null-space Projection module to eliminate non-linear topic leakage.
- Generated stratified data buckets and interactive review sheets for 420+ synthetic instruction samples; achieved 83% held-out corruption-contrast accuracy with 550-dim feature fusion and reduced topic probe accuracy to chance levels (0.014) to isolate pure stylistic variance.

Gaperon-llm-as-a-judge
- Worked on a scalable evaluation system that processes JSON datasets to automatically score and analyze text using Large Language Models- 6 dimensions of quality, utilizing concurrent batch processing to handle data efficiently.
- Deployed open-source models (Llama 3.2) locally via Ollama and routed them through an OpenAI-compatible API endpoint, enabling secure, zero-cost, and privacy-preserving model inference.

Multilingual LLM Evaluation Pipeline
- Worked on a Kubernetes-based evaluation infrastructure (V100 GPU) for multilingual LLM assessment using SkyPilot orchestration; debugged and resolved backend issues.
- Generated technical analysis report and interactive HTML visualization for 4,900+ evaluation samples on Hindi/Tamil MCQ reasoning: achieved 69.95% accuracy (27B) with 2.34x scaling coefficient and identified 8.33pp language-specific performance gap.

Gaperon mmbert distill repository
- Worked on a project using jhu-clsp/mmBERT-base with 6 independent classification heads to predict text quality (clarity, coherence, depth, grammar, usefulness, overall) across English, Hindi, and Tamil. Optimised the model to include datasets for languages like Bengali, Telugu and Malayalam.
- Set up end-to-end pipeline using SkyPilot with HuggingFace Hub integration for dataset streaming, checkpoint versioning, and model promotion across branches

Synthia - Synthetic Data Generation Pipeline
- Worked on a Python pipeline for generating instruction fine-tuning data (context, instruction, response triples) from raw text documents using self-hosted LLMs
- Set up and validated the local development environment, connecting to MIRA's private GPU inference cluster over Tailscale VPN.
- Built language-normalized data generation pipeline with automated quality filtering (98.9% pass rate) supporting both pure and code-mixed (Hinglish/Tanglish) outputs for real-world applicability.

Summer Intern, S3 Labs, IIT Bhiliai | May 2025- July 2025

1. **Technical Image Understanding**
   - Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferenced using LLM's..
2. **Legal Infographic Generation**
   Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
   - Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
   - Generated trial prototypes for generating Infographics like Timelines, Key Arguments
   - **Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering**

Coder's High Volunteer :
Volunteered for Workshop Series by Sudarshan Iyenge, IIT Ropar

COVID-19 Response Volunteer:
Caregiver Saathi, May 2021

Others

.GATE DA 2025: Qualified
(March 2025)

Winner Hack Kochi #2:
(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
Subject Topper for Maths, Chemistry and Computer Science
(May 2018)

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Projects

1. **Forecasting the Total Marine Fish Production In Kerala - Development & Comparative Analysis of Machine Learning Models**
Machine Learning, Fisheries | Jan 2022- May 2022
- Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
- Highest predictive power - Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07

2. **Chat Application, Publisher Subscriber Architecture | November 2019**
- Co-Developed a real-time communication system using the MQTT protocol, applying a publisher–subscriber model to support chat-style applications as part of Hack-Kochi#2

3. **VR Industrial Training Simulator | Unity, C# | Jan 2021**
Co-developed a Virtual Reality application simulating boiler feedwater plant control room operations, creating a safe, immersive environment for industrial operator training
Collaborated on engineering interactive 3D scenarios and automation workflows in Unity/C#, effectively eliminating the physical risks and costs associated with traditional models.

4. . **Prediction of Crime Categories | Ensembled Learning, Law & Order | August 2024**
- The dataset consisted of 20,000 rows & 20 columns about location, victim demographics, date, time, etc. 6 crime categories were possible for the target variable.
- Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
- **LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58

5. **Perspective Aware Sentiment Analysis| Applied Data Analytics | November 2025**
Models: Roberta-base, GPT-4o-mini
- Real-time data pipeline using **Google News RSS** to fetch and parse headlines based on specific search queries and time windows.
- Implemented a standard sentiment baseline using the **cardiff nlp/twitter-roberta-base**-sentiment-latest model to provide objective scores.
Perspective-aware sentiment engine was created using GPT-4o-mini that adapts its analysis based on user-defined personas.
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6. **Opacus - Differential Privacy in PyTorch**
Open Source Contribution, Data Privacy and Data Quality Project | May 2026
- Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes. Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler)

7.. **LLM Distillation for Named Entity Recognition |**
Applied Machine Learning Project | May 2026
Collaborated with a teammate to develop a knowledge distillation pipeline, compressing a 12B-parameter LLM (Gemma 3) into a 270M-parameter model for fine-grained NER.

8 . **Dynamic Reflections: Probing Video Representations with Text Alignment | Introduction to Generative AI| June 2026**
- Partnered with a teammate to evaluate and present cutting-edge research on multimodal video-text representation alignment and spatio-temporal semantics for the "Introduction to Generative Intelligence" course at CMI.

Education.

Chennai Mathematical Institute,
Data Science, Aug 2024 - May 2026

Master of Science